NYCU Solid State Physics

Question Collection

Editorial note

This collection reconstructs supplied Solid State Theory and Solid State Physics examination pages. Each chapter preserves the printed course title, date, question numbers, and weights. These papers come from different courses and years; this is an independent study archive rather than an official course publication.

Legible printed questions are transcribed in English. Handwritten answers and grading marks are treated as reference material, not as part of the questions. Two source groups have missing or cropped material: only the first printed page of the May 2014 examination is available, and another fragment has no visible date or initial problem setup. Their coverage is explicitly identified. Missing question text is not invented. Printed weights are retained even when they do not sum to 100%.

Read this volume alongside the separate *Detailed Solutions*. All equations and reconstructed figures have editable LaTeX sources.

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Chapter 1

Solid State Theory: Midterm, 6 May 2014

Only the printed page containing Questions 1–3 survives in the supplied material. Its printed weights total 60%. The missing printed continuation cannot be reconstructed verbatim from handwritten answers. Supplementary exercises derived from those notes appear separately in the solutions book and are not presented as recovered exam questions. Handwritten marks over the printed equations have not been incorporated.

1. (10%) True or false. Explain for your answer.

(a) (5%) Two atoms of the same type in a crystal must be connected by a translation vector of the underlying Bravais lattice.

(b) (5%) Consider a $\boldsymbol{k}$ point lying outside of the first Brillouin zone of a Bravais lattice. There must exist a reciprocal lattice point to which the $\boldsymbol{k}$ point is closer than to the origin.

2. (20%) Lattice planes

Consider a Bravais lattice spanned by the primitive vectors

$$\boldsymbol{t}_1 = \frac{a}{2}(0,1,1), \qquad \boldsymbol{t}_2 = \frac{a}{2}(1,0,1), \qquad \boldsymbol{t}_3 = \frac{a}{2}(1,1,0).$$

(a) (5%) Consider a family of direct lattice planes that is normal to $\frac{1}{2}\boldsymbol{g}_1 + \boldsymbol{g}_2 - \boldsymbol{g}_3$. Find the distance between two successive lattice planes.

(b) (10%) The lattice points on each lattice plane in (a) form a two-dimensional lattice. Find the density of lattice points (i.e., the number of lattice points per unit area) on the two-dimensional lattice plane.

(c) (5%) Consider a family of direct lattice planes that is normal to $\boldsymbol{t}_1 + 2\boldsymbol{t}_2 - 2\boldsymbol{t}_3$. Find the distance between two successive lattice planes.

Notation used in the solutions: $\boldsymbol{g}_j$ are the reciprocal primitive vectors, with $\boldsymbol{t}_i \cdot \boldsymbol{g}_j = 2\pi\delta_{ij}$. This definition is implicit on the surviving printed page.

3. (30%) Two-dimensional graphite

Consider the two-dimensional crystal of graphite, which has two carbon atoms per unit cell at positions

$$\boldsymbol{d}_1 = 0, \qquad \boldsymbol{d}_2 = a\left(0, \frac{1}{\sqrt{3}}\right).$$

The primitive translation vectors for the underlying two-dimensional hexagonal lattice are

$$\boldsymbol{t}_1 = a\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad \boldsymbol{t}_2 = a\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

Graphite has two π bands with the dispersion relations

$$E_{\boldsymbol{k}} = E_p \pm \gamma \left|F_{\boldsymbol{k}}\right|, \qquad \gamma > 0,$$

where

$$\left|F_{\boldsymbol{k}}\right|^2 = 1 + 4\cos\frac{k_x a}{2}\cos\frac{\sqrt{3}k_y a}{2} + 4\cos^2\frac{k_x a}{2}.$$

(a) (15%) Identify if or not symmetry points Γ ($\boldsymbol{k} = 0$), Q [$\boldsymbol{k} = (2\pi/a)(1/2, 1/(2\sqrt{3}))$], and P [$\boldsymbol{k} = (2\pi/a)(2/3, 0)$] of the first Brillouin zone are critical points. What type of critical point are they?

(b) (5%) Is there a gap between the two π bands?

(c) (10%) Express the density of states contributed from the states in a small region containing a critical point you choose as you like from those in (a).

Chapter 2

Solid State Theory: Midterm, 5 December 2014

Both printed pages survive. The printed question weights sum to 110%; they are retained without renormalization. Handwritten answers and annotations are not part of the transcribed questions.

1. (40%) Explain the following terms.

(a) Bravais lattices;

(b) composite crystals;

(c) Brillouin zone;

(d) reciprocal lattices;

(e) Bloch theorem;

(f) exchange energy of a many-electron system;

(g) fully spin-polarized ground state;

(h) Wigner crystallization.

2. (20%) Lattice planes and periodicity

Consider a Bravais lattice spanned by the primitive vectors

$$\boldsymbol{t}_1 = \frac{a}{2}(0,1,1), \qquad \boldsymbol{t}_2 = \frac{a}{2}(1,0,1), \qquad \boldsymbol{t}_3 = \frac{a}{2}(1,1,0).$$

$\boldsymbol{g}_1$, $\boldsymbol{g}_2$, and $\boldsymbol{g}_3$ are the primitive vectors of the reciprocal lattice, defined by the relations $\boldsymbol{t}_i \cdot \boldsymbol{g}_j = 2\pi\delta_{ij}$.

(a) (5%) Consider a family of direct lattice planes that is normal to $3\boldsymbol{g}_1 + 6\boldsymbol{g}_2 - 6\boldsymbol{g}_3$. Find the distance between two successive lattice planes.

(b) (5%) The lattice points on each lattice plane in (a) form a two-dimensional lattice. Find the density of lattice points (i.e., the number of lattice points per unit area) on the two-dimensional lattice plane.

(c) (5%) Consider a family of direct lattice planes that is normal to $\frac{1}{2}\boldsymbol{t}_1 + \boldsymbol{t}_2 - \boldsymbol{t}_3$. Find the distance between two successive lattice planes.

(d) (5%) Does the plane wave $\exp(\mathrm{i}\frac{1}{2}\boldsymbol{g}_1 \cdot \boldsymbol{r})$ have the lattice periodicity? Explain for your answer.

3. (20%) Band critical points

Consider a two-dimensional crystal that is underlain by a two-dimensional hexagonal lattice spanned by the primitive vectors

$$\boldsymbol{t}_1 = a\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad \boldsymbol{t}_2 = a\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

The crystal has two bands with the dispersion relations

$$E_{\boldsymbol{k}} = E_p \pm \gamma \left|F_{\boldsymbol{k}}\right|, \qquad \gamma > 0,$$

where

$$|F_{\boldsymbol{k}}|^2 = 1 + 4\cos\frac{k_x a}{2}\cos\frac{\sqrt{3}k_y a}{2} + 4\cos^2\frac{k_x a}{2}.$$

(a) (10%) Identify if or not symmetry points Γ ($\boldsymbol{k} = 0$) and P [$\boldsymbol{k} = (2\pi/a)(2/3, 0)$] of the first Brillouin zone are critical points. What type of critical point are they if they are critical points?

(b) (10%) Express the density of states contributed from the states in a small region containing a critical point you choose as you like from those in (a).

4. (5%) Many-electron Hamiltonian

Express the Hamiltonian for a system of N interacting electrons in the presence of nuclei fixed in a configuration.

5. (5%) Hartree trial state

Describe the trial wavefunction for the ground state of a many-electron system in the Hartree theory.

6. (10%) Hartree–Fock trial state

Describe the trial wavefunction for the ground state of a many-electron system in the Hartree–Fock theory. How to obtain the approximate ground-state wavefunction?

7. (10%) Monte Carlo trial state

Describe the trial wavefunction in the Monte Carlo Method of Ceperley and Alder for the ground state of a many-electron system in a uniform neutralizing background of positive charges.

Chapter 3

Solid State Physics: Midterm Exam, April 30, 2015

Printed total: 100%.

Reconstructed from two pages of the source exam, retaining question numbers, marks, and English question content. Handwritten annotations are excluded from the question text.

1. True or false (15%)

True or false. Explain your answer. Crystalline $BaTiO_3$ is underlain by a simple cubic lattice with a basis consisting of five atoms: a Ba atom, a Ti, and three O atoms.

(a) (5%) A vector connecting two Ti atoms must be a translation vector of the lattice.

(b) (5%) A vector connecting a Ba atom with a Ti atom must not be a translation vector of the lattice.

(c) (5%) A vector connecting two O atoms must be a translation vector of the lattice.

2. Direct and reciprocal lattices (30%)

Consider a Bravais lattice spanned by the primitive vectors

$$\boldsymbol{t}_1 = \frac{a}{2}(0,1,1), \qquad \boldsymbol{t}_2 = \frac{a}{2}(1,0,1), \qquad \boldsymbol{t}_3 = \frac{a}{2}(1,1,0).$$

(a) (10%) Find the volumes of the Wigner–Seitz cell and the first Brillouin zone.

(b) (5%) Find the three primitive vectors $\boldsymbol{g}_1, \boldsymbol{g}_2, \boldsymbol{g}_3$ of the reciprocal lattice, satisfying $\boldsymbol{t}_i \cdot \boldsymbol{g}_j = 2\pi\delta_{ij}$, $i, j = 1, 2, 3$.

(c) (5%) Consider a family of direct lattice planes that is normal to $\boldsymbol{t}_1$. Find the distance between two successive lattice planes.

(d) (5%) The lattice points on each lattice plane normal to $\boldsymbol{t}_1$ form a two-dimensional lattice. Find the density of lattice points (i.e., the number of lattice points per unit area) on the two-dimensional lattice plane.

(e) (5%) Can the three vectors

$$\boldsymbol{G}_1 = \frac{\pi}{a}(0,1,1), \qquad \boldsymbol{G}_2 = \frac{\pi}{a}(1,0,1), \qquad \boldsymbol{G}_3 = \frac{\pi}{a}(1,1,0)$$

span the reciprocal lattice? Explain your answer.

3. Diamond structure (5%)

A crystal in diamond structure has lattice constant a. Find the density of atoms (i.e., the number of atoms per unit volume).

4. Definitions (20%)

Describe the following terms:

(a) (5%) three-dimensional hexagonal lattice;

(b) (5%) Bloch theorem;

(c) (5%) critical point;

(d) (5%) Fermi energy.

5. Critical point and density of states (10%)

Consider a three-dimensional crystal with a critical point $\boldsymbol{k}_1$ of type M_1.

(a) (5%) Express the energy dispersion for $\boldsymbol{k}$ in a small neighborhood of the critical point.

(b) (5%) Find the density of states (as a function of energy) for the states in the neighborhood.

6. Density parameter of a free-electron gas (10%)

Consider a three-dimensional free-electron gas with a dimensionless parameter r_s for its electron density. Express in terms of r_s the electron density n, the Fermi wavenumber k_F, the Fermi energy ε_F, the Fermi temperature T_F, and the average electron energy for the gas at the ground state.

7. Electronic heat capacity (5%)

Consider a three-dimensional gas of N free electrons. Explain that the gas has a heat capacity

$$C_V \approx \frac{3}{2} k_\mathrm{B} N \frac{T}{T_F}$$

rather than the classical $C_V = \frac{3}{2} k_\mathrm{B} N$.

> The estimate is retained as printed. The solutions explain why its numerical coefficient differs from the rigorous low-temperature expansion.

8. Fermi–Dirac tail (5%)

Explain why the Fermi–Dirac distribution function in a simple metal can be approximated well as an exponential function for electrons with energy higher than the vacuum level.

Chapter 4

Solid State Physics: Final Exam, June 25, 2015

Printed total: 100%.

> Reconstructed from one complete exam page. Two identical source photographs are represented by this single exam. Handwritten answers are excluded.

1. Bloch sums (15%)

Express a Bloch sum of wave vector $\boldsymbol{k}$. Show that the Bloch sum has the property of the Bloch function. What is the role the Bloch sums play in the tight-binding method?

2. Tight-binding model of silicon (15%)

The electronic configuration of the silicon atom is $1s^2 2s^2 2p^6 3s^2 3p^2$. If five orbitals for each atom are chosen to construct the Bloch sums in the tight-binding method of band structure calculation for electrical and chemical properties of crystalline silicon, what is your suggestion on the five orbitals? How many bands will you have? How many bands will be fully filled by valence electrons for the silicon crystal in the ground state?

3. Plane-wave expansion (10%)

Describe the difficulty in the band structure calculation when plane waves are used as a basis to expand the crystal wavefunction. How to overcome the difficulty?

4. Pseudopotential (10%)

Explain the term "pseudopotential". Why can the pseudo wavefunctions for a crystal be expanded with a reasonable number of plane waves?

5. Adiabatic energy surface (5%)

Explain the term "adiabatic energy surface".

6. Normal modes from an energy surface (15%)

Consider an electronic-nuclear crystal system on a nondegenerate ground adiabatic surface $E_0(\boldsymbol{R})$ which well separates in energy from all the other adiabatic surfaces. Describe how to obtain the normal modes of the lattice dynamics with the information of the adiabatic surface $E_0(\boldsymbol{R})$ and the atomic masses.

7. Dynamical matrix (5%)

Express the dynamical matrix elements, given the atomic masses M_ν and the force constants $D_{\nu\alpha,\nu'\alpha'}(\boldsymbol{t}_n)$, where ν, ν' are indices for basis atoms, α, α' are indices for Cartesian coordinate components x, y, z, and $\boldsymbol{t}_n$ stands for translation vectors.

8. Lattice dynamics of $BaTiO_3$ (20%)

Consider the $BaTiO_3$ crystal with N unit cells, which has a Ba atom, a Ti atom, and three O atoms in each unit cell.

(a) What is the dimension of the dynamical matrix?

(b) How many normal modes are there of the lattice dynamics?

(c) How many acoustic branches are there?

(d) How many optical branches are there?

9. Optical-mode polarization vectors (5%)

Consider a crystal with two atoms in each unit cell. The two atoms have masses M_1 and M_2, respectively. Find the value of

$$\frac{|\boldsymbol{A}_1|^2}{\boldsymbol{A}_1^* \cdot \boldsymbol{A}_2},$$

where $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are the polarization vectors of the optical phonon modes in the long-wavelength limit $(q \to 0)$.

Hint: The atomic displacements of mode $(\boldsymbol{q}, p)$ may be expressed by

$$\boldsymbol{u}_{n\nu}^{(\boldsymbol{q},p)}(t) = \frac{\boldsymbol{A}_\nu(\boldsymbol{q},p)}{\sqrt{M_\nu}} w_{\boldsymbol{q}n} \mathrm{e}^{-\mathrm{i}\omega(\boldsymbol{q},p)t}.$$

Chapter 5

Solid State Theory: Midterm, 29 April 2016

The supplied printed sheet contains all twelve questions, whose weights total 100%. Mathematical subscripts and line breaks have been typeset consistently; handwritten answers are excluded.

1. (15%) True or false. Explain for your answer.

(a) (5%) The determinantal wavefunction with spin-orbitals satisfying the Hartree–Fock equations is the exact solution of the Schrödinger equation for many interacting electrons.

(b) (5%) Consider a homogeneous electron gas embedded in a uniform positive charge background neutralizing the negative charge of the electron gas. The electron gas tends to be fully spin-polarized as the electron density increases.

(c) (5%) A crystal with an odd number of electrons in each primitive cell cannot be an electrical insulator.

2. (5%) Hamiltonian

Express the Hamiltonian for a system of N interacting electrons in the presence of nuclei fixed in a configuration. Explain each term in your expression.

3. (10%) Trial wave functions

What are the trial wave functions for the many-electron system in Problem 2 in the Hartree theory and in the Hartree–Fock theory, respectively?

4. (10%) Hartree–Fock equations

Express the Hartree–Fock equations and explain each term in your expression. Roughly describe how the equations result from the trial wave functions.

5. (5%) Hohenberg–Kohn functional

Express and explain the Hohenberg–Kohn functional.

6. (10%) Kohn–Sham equations

Express the Kohn–Sham equations and explain each term in your expression. Roughly describe how the equations result from the Hohenberg–Kohn functional.

7. (5%) Variational collapse

Describe and explain the variational collapse in the expansion of crystal states with pure plane waves.

8. (5%) Modified plane waves

How to modify the basis functions of plane waves to cure the variational collapse problem?

9. (5%) Pseudo wavefunction

Express the relation between a crystal state wavefunction $\psi_{\boldsymbol{k}}$ and its pseudo wavefunction $\varphi_{\boldsymbol{k}}$.

10. (10%) Effective Hamiltonian

The Schrödinger equation for a crystal state $\psi_{\boldsymbol{k}}$ can become an effective Schrödinger equation for the pseudo wavefunction $\varphi_{\boldsymbol{k}}$ with an effective Hamiltonian $H^{(\mathrm{eff})}$.

(a) (5%) Express and explain the effective Hamiltonian $H^{(\mathrm{eff})}$.

(b) (5%) Roughly describe how to solve the effective Schrödinger equation using the basis of plane waves.

11. (5%) Sodium valence energy

Conventionally, the $3s$ valence electron of the sodium atom is regarded to experience an effective Coulomb potential of a point-like charge, as the $1s$ electron of the hydrogen atom. Explain the fact that the $3s$ orbital energy of the sodium atom is about $-5.14\,\mathrm{eV}$ instead of $-13.6\,\mathrm{eV}$ for the $1s$ orbital energy of the hydrogen atom.

12. (5%) Pseudo atom

Explain the term “pseudo atom.”

Chapter 6

Solid State Theory: Midterm Exam, April 26, 2017

Reconstructed from two printed question pages; the marks total 100%. Handwritten answers accompanying the source are not part of the questions and are not reproduced. Vectors are consistently typeset in bold.

1. (10%) True or false. Explain for your answer.

(a) (5%) Two atoms of the same type in a crystal must be connected by a translation vector of the underlying Bravais lattice.

(b) (5%) Consider a $\mathbf{k}$ point lying outside of the first Brillouin zone of a Bravais lattice. There must exist a reciprocal lattice point to which the $\mathbf{k}$ point is closer than to the origin.

2. (20%)

Consider a Bravais lattice spanned by the primitive vectors

$$\mathbf{t}_1 = \frac{a}{2}(0,1,1), \qquad \mathbf{t}_2 = \frac{a}{2}(1,0,1), \qquad \mathbf{t}_3 = \frac{a}{2}(1,1,0).$$

(a) (5%) Consider a family of direct lattice planes that is normal to $\frac{1}{2}\mathbf{g}_1 + \mathbf{g}_2 - \mathbf{g}_3$. Find the distance between two successive lattice planes.

(b) (10%) The lattice points on each lattice plane in (a) form a two-dimensional lattice. Find the density of lattice points (i.e., the number of lattice points per unit area) on the two-dimensional lattice plane.

(c) (5%) Consider a family of direct lattice planes that is normal to $\mathbf{t}_1 + 2\mathbf{t}_2 - 2\mathbf{t}_3$. Find the distance between two successive lattice planes.

3. (25%)

Consider the two-dimensional crystal of graphite, which has two carbon atoms per unit cell at positions

$$\mathbf{d}_1 = 0, \qquad \mathbf{d}_2 = a(0, 1/\sqrt{3}).$$

The primitive translation vectors for the underlying two-dimensional hexagonal lattice are

$$\mathbf{t}_1 = a(1/2, \sqrt{3}/2), \qquad \mathbf{t}_2 = a(-1/2, \sqrt{3}/2).$$

Graphite has two π bands with dispersion

$$\epsilon_{\mathbf{k}} = \epsilon_p \pm \gamma |F_{\mathbf{k}}|,$$

where ϵ_p is a constant, $\gamma > 0$, and

$$|F_{\mathbf{k}}|^2 = 1 + 4\cos(k_x a/2)\cos(\sqrt{3}k_y a/2) + 4\cos^2(k_x a/2).$$

(a) (15%) Identify if or not symmetry points Γ ($\mathbf{k} = 0$), Q [$\mathbf{k} = (2\pi/a)(1/2, 1/(2\sqrt{3}))$], and P [$\mathbf{k} = (2\pi/a)(2/3, 0)$] of the first Brillouin zone are critical points. What type of critical point are they?

(b) (10%) Express the density of states contributed from the states in a small region containing a critical point you choose as you like from those in (a).

4. (5%)

Describe the Bloch theorem.

5. (5%)

Explain the negative value of the exchange energy in a many-electron system from the viewpoint of the difference between the Hartree-Fock and the Hartree theories.

6. (10%)

Describe the Hohenberg-Kohn functional.

7. (10%)

Describe the local density approximation (LDA) in the exchange-correlation functional.

8. (15%)

Write down the Hartree, the Hartree-Fock, and the Kohn-Sham equations and explain each term in the equations. Briefly describe how to solve the equations.

Chapter 7

Solid State Physics: Midterm Exam, November 8, 2017

Printed total: 100%.

> Reconstructed from two complete exam pages. The two-atoms-per-primitive-cell premise in Question 3 and the estimate in Question 9(e) are retained as printed; their interpretation is addressed in the solutions.

1. Primitive vectors of the same lattice (5%)

Find the value of α if two sets of primitive vectors $\boldsymbol{t}_1, \boldsymbol{t}_2, \boldsymbol{t}_3$ and $\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3$ span the same Bravais lattice and

$$\begin{aligned} \boldsymbol{s}_1 &= 2\boldsymbol{t}_1 - \boldsymbol{t}_2 + 2\boldsymbol{t}_3, \\ \boldsymbol{s}_2 &= -\boldsymbol{t}_1 + \boldsymbol{t}_2 + \alpha\boldsymbol{t}_3, \\ \boldsymbol{s}_3 &= \boldsymbol{t}_1 + \alpha\boldsymbol{t}_2 + \boldsymbol{t}_3. \end{aligned}$$

2. Silicon crystal (5%)

Describe silicon crystal in terms of the Bravais lattice and basis vectors.

3. Body-centered cubic lattice (20%)

An elemental crystal is formed in the body-centered cubic lattice with lattice constant a. There are two atoms in each primitive cell.

(a) (5%) What is macroscopically the volume density of atoms, i.e., the number of atoms per unit volume?

(b) (5%) What is the volume of the Wigner–Seitz cell of the bcc lattice?

(c) (5%) What is the volume of the Brillouin zone for the crystal?

(d) (5%) How many facets does the Brillouin zone have? Explain your answer.

4. Reciprocal lattice and plane density (10%)

Consider a Bravais lattice spanned by the primitive vectors

$$\boldsymbol{t}_1 = \frac{a}{2}(0,1,1), \qquad \boldsymbol{t}_2 = \frac{a}{2}(1,0,1), \qquad \boldsymbol{t}_3 = \frac{a}{2}(1,1,0).$$

(a) (5%) Find the three primitive vectors $\boldsymbol{g}_1, \boldsymbol{g}_2, \boldsymbol{g}_3$ that span the reciprocal lattice and satisfy $\boldsymbol{t}_i \cdot \boldsymbol{g}_j = 2\pi\delta_{ij}$, $i, j = 1, 2, 3$.

(b) (5%) Consider a family of direct lattice planes with a distance d between two successive lattice planes. Find the density of lattice points (number of lattice points per unit area) of each of the lattice planes.

5. Born–von Karman sum (5%)

Evaluate the sum

$$\frac{1}{V}\sum_{\boldsymbol{k}} k_x k_y k_z \exp\left(-\frac{k_x^2}{\alpha} - \frac{k_y^2}{\beta} - \frac{k_z^2}{\gamma}\right)$$

over all $\boldsymbol{k}$ allowed by the Born–von Karman cyclic boundary condition, where V is the system volume and $\alpha, \beta, \gamma > 0$.

6. Bloch translation (5%)

If a crystal wavefunction indexed by $\boldsymbol{k}$ obeying the Bloch theorem is $f(\boldsymbol{r})$ in the Wigner–Seitz cell containing lattice point $\boldsymbol{t}_n$, what is the wavefunction in the Wigner–Seitz cell containing lattice point $\boldsymbol{t}_m$?

7. Anisotropic dispersion (12%)

Consider a band of dispersion

$$\varepsilon_{\boldsymbol{k}} = \varepsilon_0 + \frac{(k_x - k_0)^2}{\alpha} + \frac{k_y^2}{\beta} + \frac{k_z^2}{\gamma}, \qquad \alpha, \beta, \gamma > 0,$$

for a crystal of volume V.

(a) (6%) Where is the critical point? What type is the critical point?

(b) (6%) Find the density of states for the band.

8. Free-electron gas in an fcc metal (12%)

Consider an electron gas in a simple crystal of metal composed of atoms of each with three valence electrons when isolated. The crystal is in the face-centered cubic lattice with lattice constant a. The electron gas consists of the electrons originating from all the valence electrons of atoms, and can be regarded as a free-electron gas. What are r_s, k_F, and ε_F, and the average electron energy at $T = 0$?

9. Sommerfeld expansion (26%)

The Sommerfeld expansion to order $(T/T_F)^2$,

$$\int_{-\infty}^{\infty} \Gamma(\varepsilon) f(\varepsilon)\, \mathrm{d}\varepsilon = \int_{-\infty}^{\mu} \Gamma(\varepsilon)\, \mathrm{d}\varepsilon + \frac{\pi^2}{6}(k_\mathrm{B}T)^2 \Gamma'(\mu),$$

is useful for solving the following problems.

(a) (6%) Show that the change in the chemical potential μ of a free-electron gas with temperature T satisfies

$$\frac{\mathrm{d}\mu}{\mathrm{d}T} = -\frac{\pi^2}{3} k_\mathrm{B}^2 T \frac{D'(\mu)}{D(\mu)},$$

where D is the density of states.

(b) (5%) From (a), show that for a three-dimensional free-electron gas,

$$\mu(T) = \mu_0 \left[1 - \frac{\pi^2}{12}(T/T_F)^2\right],$$

where μ_0 is the chemical potential at $T = 0$ (i.e., the Fermi energy).

(c) (5%) Show that the internal energy of the electron gas is

$$U(T) = \int_{-\infty}^{\mu} \varepsilon D(\varepsilon)\, \mathrm{d}\varepsilon + \frac{\pi^2}{6}(k_\mathrm{B}T)^2 [D(\mu) + \mu D'(\mu)].$$

(d) (5%) Show that the electronic heat capacity at constant volume is

$$C_V(T) = \frac{\pi^2}{3} k_\mathrm{B}^2 T D(\mu_0).$$

(e) (5%) Explain that the electron gas has a heat capacity $C_V \approx \frac{3}{2} k_\mathrm{B} N \frac{T}{T_F}$ rather than the classical value $\frac{3}{2} k_\mathrm{B} N$, where N is the number of electrons.

Chapter 8

Solid State Theory: Final, June 24, 2020

The supplied printed page contains Questions 1–12. Its visible weights sum to 85%; they are retained below. Handwritten responses are excluded from the transcription.

1. (5%) Describe the main differences between the Hartree theory and the Hartree–Fock theory.

2. (10%) From the expression of the energy of the normal ground state of the homogeneous electron gas in the Hartree–Fock approximation

$$E_0^{(\mathrm{N})} = N\left(\frac{3}{5}\frac{\hbar^2 k_F^2}{2m} - \frac{3}{4}\frac{e^2 k_F}{\pi}\right),$$

express the energy of the fully spin polarized ground state and explain your answer.

3. (10%) Write down the Hartree and the Hartree–Fock equations and explain each term in these equations. Describe how to solve the equations.

4. (5%) Describe the advantage of the density functional theory over the Hartree–Fock theory.

5. (5%) Express and explain the Hohenberg–Kohn functional.

6. (5%) Express and explain the Kohn–Sham equations.

7. (5%) Describe the connection of the Kohn–Sham equations with the Hohenberg–Kohn functional.

8. (5%) Given $\varepsilon_{\mathrm{xc}}(n)$, derive an expression for $V_{\mathrm{xc}}^{(\mathrm{LDA})}(\boldsymbol{r})$, where $\varepsilon_{\mathrm{xc}}(n)$ is the many-electron exchange-correlation energy per electron of a homogeneous interacting electron gas of electron density n, and $V_{\mathrm{xc}}^{(\mathrm{LDA})}(\boldsymbol{r})$ is the exchange-correlation potential in the local density approximation.

9. (5%) Describe the Bloch sums in the tight-binding method for the band structure calculation of solids.

10. (15%) The electronic configuration of the silicon atom is $1s^2 2s^2 2p^6 3s^2 3p^2$.

(a) (5%) If five orbitals for each atom are chosen to construct the Bloch sums in the tight-binding method of band structure calculation for electrical and chemical properties of crystalline silicon, what is your suggestion on the five orbitals?

(b) (5%) From (a), how many bands will you have?

(c) (5%) From (b), how many bands will be fully filled by valence electrons for the silicon crystal in the ground state?

11. (10%) What is the difficulty in band structure calculation of solids using plane waves as basis functions? How to solve the difficulty?

12. (5%) Describe the core Bloch sum(s) in the orthogonalized plane waves for calculation of the band structure of crystal lithium hydride (LiH).

Chapter 9

Solid State Theory: Midterm Exam, April 23, 2021

Reconstructed from two printed question pages; the marks total 100%. Six accompanying pages of cover material and handwritten work are not reproduced. Question 4 retains the printed formula for $|F_{\mathbf{k}}|^2$, without incorporating the nearby handwritten alteration.

1. (45%) True or false. Explain for your answer.

Each item is worth 3%.

(a) The sites of carbon atoms in a diamond crystal form a Bravais lattice.

(b) Consider two sets of vectors in two-dimensional space,

$$\{\mathbf{t}_1 = (2a, -b),\ \mathbf{t}_2 = (-a, b)\} \quad \text{and} \quad \{\mathbf{s}_1 = (a, 3b),\ \mathbf{s}_2 = (-2a, -b)\}.$$

The two sets of vectors are the primitive vectors spanning the same Bravais lattice.

(c) Two different Bravais lattices having the same point group belong to the same crystal system.

(d) Two atoms of the same type in a simple crystal must be connected by a translation vector of the underlying Bravais lattice.

(e) For all three-dimensional Bravais lattices, the product of the Wigner-Seitz cell and the Brillouin zone is constant.

(f) According to the Born-von Karman boundary condition, the number of allowed **k** points in the Brillouin zone is equal to the number of the primitive cells in the crystal. Therefore, the density of **k** points is proportional to the crystal volume.

(g) According to the Bloch theorem, an eigenfunction of the crystal Hamiltonian in a unit cell can be determined from that in one of other unit cells.

(h) For a many-electron system, the ground-state energy in the Hartree-Fock theory must be lower than that in the Hartree theory.

(i) The determinantal wavefunction formed with the spin-orbitals that satisfy the Hartree-Fock equations is the eigenfunction of the interacting many-electron Hamiltonian.

(j) In the Hartree-Fock theory, the determinantal wavefunction formed with the spin-orbitals that satisfy the Hartree-Fock equations is the best wavefunction for the ground state of the interacting many-electron Hamiltonian.

(k) The energy of the normal ground state of a homogeneous electron gas is always lower than that of the fully spin-polarized ground state.

(l) In the Hohenberg-Kohn functional, the electron density depends on the external potential.

(m) The Kohn-Sham orbitals form the determinantal wavefunction of the ground state of an interacting many-electron system.

(n) For an interacting many-electron ground state, the electron density can be expressed by $\sum_{i=1}^{N} |\phi_i(\mathbf{r})|^2$ and the kinetic energy can be expressed by $\sum_{i=1}^{N} \langle \phi_i | \mathbf{p}^2/(2m) | \phi_i \rangle$, where ϕ_i are the Kohn-Sham orbitals.

(o) The Kohn-Sham equations determine exactly the ground-state electron density and the electronic ground-state energy.

2. (20%) Describe and explain the following terms.

(a) (5%) Hamiltonian for a system of N interacting electrons in the presence of nuclei fixed in a configuration

(b) (5%) Hohenberg-Kohn functional

(c) (5%) Exchange-correlation functional in the density functional theory

(d) (5%) Local density approximation in the density functional theory

3. (15%)

Consider a face-centered cubic lattice of lattice constant a.

(a) (5%) What is the volume of the primitive cell? Accordingly, find the lattice point density?

(b) (5%) Write out three primitive vectors, and accordingly find the three primitive vectors spanning the reciprocal lattice.

(c) (5%) Describe the two-dimensional lattice to the plane of which the body diagonal of the conventional unit cell is normal.

4. (20%)

Consider the two-dimensional crystal of graphite, which has two carbon atoms per unit cell at positions

$$\mathbf{d}_1 = 0, \qquad \mathbf{d}_2 = a(0, 1/\sqrt{3}).$$

The primitive translation vectors for the underlying two-dimensional hexagonal lattice are

$$\mathbf{t}_1 = a(1/2, \sqrt{3}/2), \qquad \mathbf{t}_2 = a(-1/2, \sqrt{3}/2).$$

Graphite has two π bands with dispersion

$$\epsilon_{\mathbf{k}} = \epsilon_p \pm \gamma |F_{\mathbf{k}}|,$$

where ϵ_p is a constant, $\gamma > 0$, and

$$|F_{\mathbf{k}}|^2 = 1 + 4\cos(k_x a/2)\cos(\sqrt{3}k_y a/2) + 4\cos^2(k_x a/2).$$

(a) (5%) Find a set of primitive vectors of the reciprocal lattice.

(b) (10%) Identify if or not symmetry points Γ ($\mathbf{k} = 0$), Q [$\mathbf{k} = (2\pi/a)(1/2, 1/(2\sqrt{3}))$], and P [$\mathbf{k} = (2\pi/a)(2/3, 0)$] of the first Brillouin zone are critical points. What type of critical point are they?

(c) (5%) Express the density of states contributed from the states in a small region containing a critical point you choose as you like from those in (b).

Chapter 10

Solid State Theory: Undated Midterm Fragment

Two photographs preserve the end of Question 2 and Questions 3–7. The date, Question 1, and the model and parameters at the start of Question 2 are outside the photographed area. The visible weights total 85%. No missing wording has been reconstructed from guesses or handwritten answers. The source calls the single-layer honeycomb model "two-dimensional graphite"; this terminology is retained below.

2. Band gaps and effective mass: incomplete question

Only the following instruction and subparts are visible: Estimate the following quantities:

(a) (10%) The band gap between the lowest two bands.

(b) (5%) The band gap between the second and the third lowest bands.

(c) (5%) The effective mass for the states about the top of the lowest band.

The defining potential, model, and numerical parameters are missing.

3. Tight-binding dispersion (10%)

Derive the band dispersion

$$E_k = E_{k=0} + 2\gamma(\cos ka - 1)$$

according to the tight-binding model, where a is the lattice constant and $\gamma < 0$ is the inter-orbital overlap (the hopping parameter).

4. True or false (10%)

Explain your answer.

(a) (5%) Two atoms of the same type in a crystal must be connected by a translation vector of the underlying Bravais lattice.

(b) (5%) Consider a $\boldsymbol{k}$ point lying outside the first Brillouin zone of a Bravais lattice. There must exist a reciprocal lattice point to which the $\boldsymbol{k}$ point is closer than to the origin.

5. Lattice planes (15%)

Consider a Bravais lattice spanned by the primitive vectors

$$\boldsymbol{t}_1 = \frac{a}{2}(0,1,1), \qquad \boldsymbol{t}_2 = \frac{a}{2}(1,0,1), \qquad \boldsymbol{t}_3 = \frac{a}{2}(1,1,0).$$

Here $\boldsymbol{g}_i$ denote the reciprocal primitive vectors.

(a) (5%) Consider a family of direct lattice planes normal to $\frac{1}{2}\boldsymbol{g}_1 + \boldsymbol{g}_2 - \boldsymbol{g}_3$. Find the distance between two successive lattice planes.

(b) (5%) The lattice points on each plane in (a) form a two-dimensional lattice. Find the number of lattice points per unit area on that plane.

(c) (5%) Consider a family of direct lattice planes normal to $\boldsymbol{t}_1 + 2\boldsymbol{t}_2 - 2\boldsymbol{t}_3$. Find the distance between two successive lattice planes.

6. Two-dimensional graphite (25%)

Consider the two-dimensional crystal of graphite, which has two carbon atoms per unit cell at positions

$$\boldsymbol{d}_1 = 0, \qquad \boldsymbol{d}_2 = a(0, 1/\sqrt{3}).$$

The primitive translation vectors for the underlying two-dimensional hexagonal lattice are

$$\boldsymbol{t}_1 = a(1/2, \sqrt{3}/2), \qquad \boldsymbol{t}_2 = a(-1/2, \sqrt{3}/2).$$

Graphite has two π bands with dispersion

$$\varepsilon_{\boldsymbol{k}} = \varepsilon_p \pm \gamma|F_{\boldsymbol{k}}|, \qquad |F_{\boldsymbol{k}}|^2 = 1 + 4\cos\frac{k_x a}{2}\cos\frac{\sqrt{3}k_y a}{2} + 4\cos^2\frac{k_x a}{2},$$

where ε_p is constant and $\gamma > 0$.

(a) (15%) Identify whether the symmetry points

$$\Gamma:\ \boldsymbol{k} = 0, \qquad Q:\ \boldsymbol{k} = \frac{2\pi}{a}\left(\frac{1}{2}, \frac{1}{2\sqrt{3}}\right), \qquad P:\ \boldsymbol{k} = \frac{2\pi}{a}\left(\frac{2}{3}, 0\right)$$

of the first Brillouin zone are critical points. What types of critical points are they?

(b) (10%) Express the density of states contributed by states in a small region containing a critical point of your choice from (a).

7. Bloch theorem (5%)

Describe the Bloch theorem.